

FIGURE 5 NEMA phase structure with protected left-turns

protected mode to omit the effect of conflict zone between the through and left-turn movements. At each side of the barrier, the left-turn movement can be served before (leading) or after (lagging) the through movement. Hence, the phase sequence decision variable ps_i can take four values of $\{1,2,3,4\}$ representing the order of left-turn phase in two sides of the barrier which are lag-lag, lead-lag, lag-lead, and lead-lead, respectively. The phase length for each of the movements p_i^c is equal to the sum of green time, yellow time, and red clearance interval which can be expressed as:

$$p_i^c = g_i^c + y_i^c + ar_i^c \quad (44)$$

3.7 | Constraints

- Cycle length

The sum of the phase lengths including green times, yellow times, and red clearance intervals of each ring is constrained to cycle length which can be expressed as:

$$\sum_{i=1}^4 p_i^c = \sum_{i=1}^4 (g_i^c + y_i^c + ar_i^c) = \Gamma \quad (45)$$

$$\sum_{i=5}^8 p_i^c = \sum_{i=5}^8 (g_i^c + y_i^c + ar_i^c) = \Gamma \quad (46)$$

- Minimum green time

The green time of the movements is required to satisfy a constraint of minimum green time $g_{min}^{l,c}$ for providing safe pedestrian passage and ensuring that no phase will be skipped. The constraint can be expressed as follows:

$$g_i^c \geq g_{min}^{l,c} \quad \forall i \in L \quad (47)$$

- Protected left-turn movements

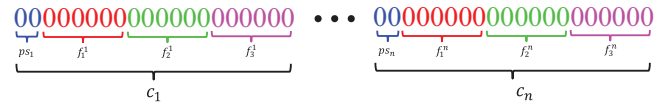


FIGURE 6 Configuration of the GA candidate solution

Due to the protected operation of the left-turn phases, the turning and through phases of the two opposite approaches are forced to start and finish at the same time on each side of the barrier. This can be expressed as:

$$p_i^c = p_{i+4}^c \quad \forall i \in [1, 2, 3, 4] \quad (48)$$

4 | GENETIC ALGORITHM OPTIMIZATION

At each KATC execution, the optimal solution for the MINLP problem is achieved by GA [44]. Due to the simple encoding scheme and high efficiency in tackling non-linearity, GA was applied to several non-linear ATSP problems [10, 18, 45]. The binary encoding scheme introduced by Park et al. [46, 47] for signal timing parameters is selected and revised to account for the protected left-turn case. The encoding scheme of a GA individual is illustrated in Figure 6. The four possible cases of phase sequence in each cycle within the decision horizon are represented by two bits. Besides, three coefficient factors f_i^c , each consisting of 6 bits, are considered to distribute the green time between the major and minor movements.

GA uses the maximization procedure in searching for the global optimum. Hence, to adapt the delay minimization problem to GA, the inverse of average passenger delay is adopted as the fitness function. Elitism is adopted in the GA optimization. Thus, before further modifications, the best solution obtained for each generation is transmitted to the next generation. The methods of mask-based crossover and adaptive mutation are implemented in this study. In the selected adaptive mutation method, the mutation probability is gradually increased with the progress of the algorithm to improve the diversity of the population, especially in final generations. Due to the unpredictable computation time in GA methods in which the solution is delivered when the algorithm fails to maintain a desirable convergence rate, in this study, a fixed number of generations is set as termination criteria.

The GA parameters are tuned based on sensitivity analysis to find the optimal values such that the computational effort of the algorithm for solving a preliminary analytical test is restricted to 10 s of CPU time on a PC with a Core i7 2.00 GHz processor with 4.00 GB RAM. Hence, the method could be potentially applied to real field control systems. The optimal parameter setting capable of simultaneously satisfying the computation time is achieved as the population size and the number of generations of 150, and the initial and final mutation probability of 0.01 and 0.02, respectively.

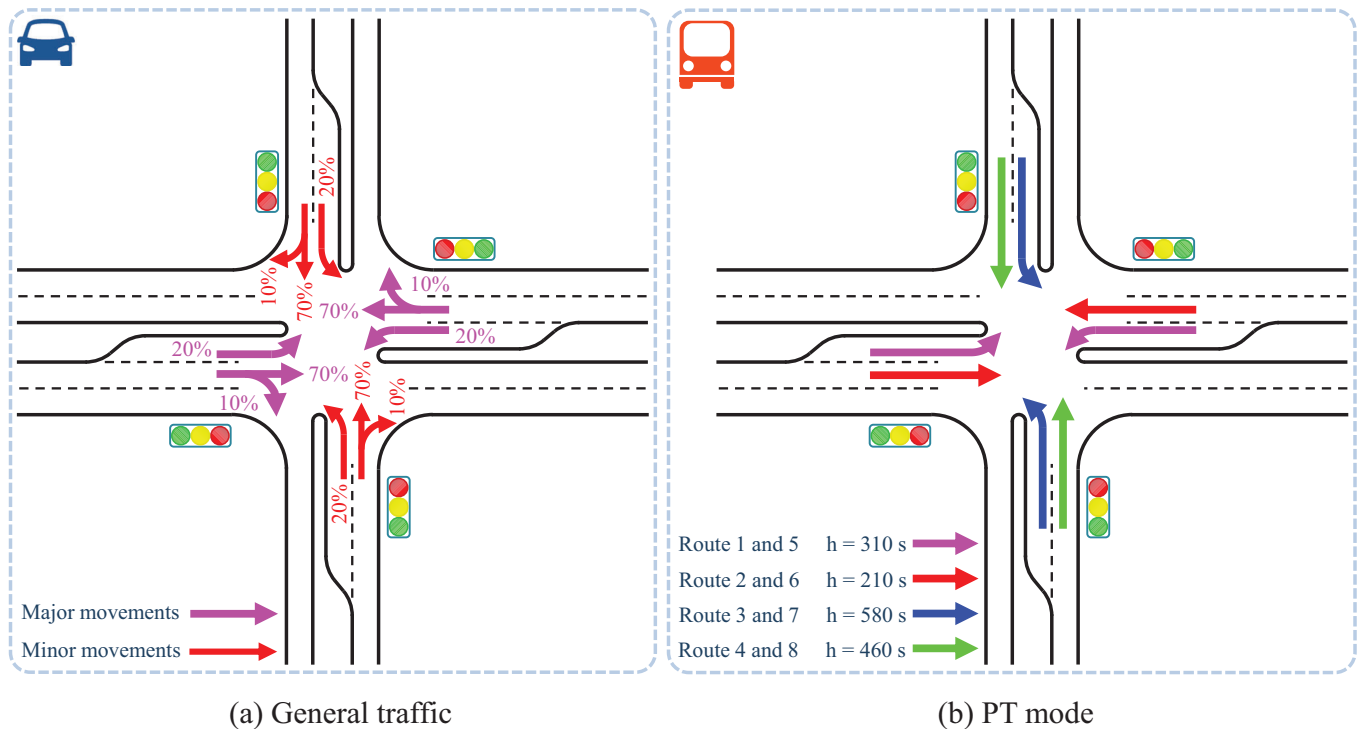


FIGURE 7 Test intersection for simulation experiments

5 | SIMULATION EXPERIMENTS

The evaluation of KATC at intersection level is performed using the VISSIM microsimulation model [48]. The core elements of the KATC algorithm including GA optimization and kinematic wave analysis are written in Python. The VISSIM COM interface is utilized to exchange data between the traffic network, control system, and the analytical components of KATC. KATC is set up to update the signal timing of the test intersection at 1-cycle intervals based on the optimal solution of the GA element over a 3-cycle decision horizon.

5.1 | Geometric configuration

The hypothetical intersection illustrated in Figure 7 consists of 2-lane major and 1-lane minor approaches. The exclusive left-turn lanes with sufficient lengths are considered for both major and minor left-turn movements. Therefore, the left-turn and through queues develop independently.

5.2 | General traffic and public transportation mode settings

The intersection undergoes a wide range of degrees of saturation (DOS) from the traffic condition with the minimum disturbance with DOS of 0.4 to the saturated condition with a DOS equal to 1.0. The general traffic demand is held constant

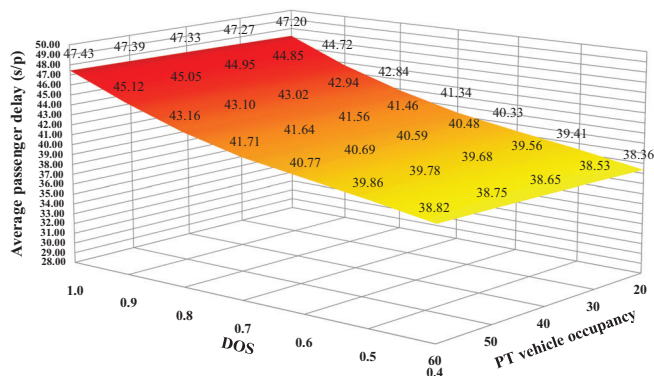
within the simulation interval. At each degree of saturation, the percentage of turning vehicles for minor and major movements are set to the values indicated in Figure 7a. The passenger occupancy of general traffic is set to 1.5. To evaluate the efficiency of KATC in handling conflicting high-demand PT movements, eight mixed-use PT routes with through and left-turn movements are generated to operate on the test intersection. For convenience, the same numeration of the NEMA phase structure is used to designate the PT routes. Headway values split by each route are shown in Figure 7b. Higher demand levels are implicitly assigned to the major approaches by assuming lower headway values for the respective routes. Various values of PT vehicle occupancy are selected to analyse the impact of the parameter on network results. The values of PT vehicle occupancy and congestion level are specified as follows:

- DOS: 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0
- PT vehicle occupancy: 20, 30, 40, 50, 60

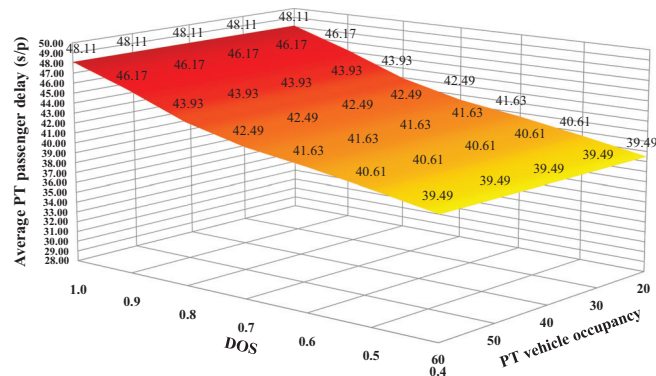
5.3 | Traffic signal controls

To evaluate the performance of the proposed model under various operating conditions, three traffic signal control models are considered as follows:

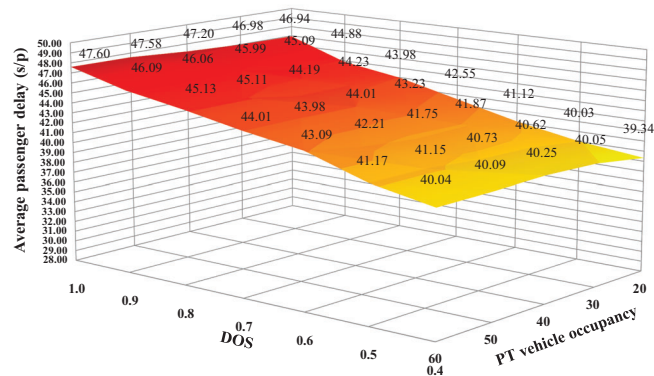
- SYNCHRO signal timing optimization model [49]
- KATC-1 (optimization of green time)
- KATC-2 (optimization of green time and phase sequence)



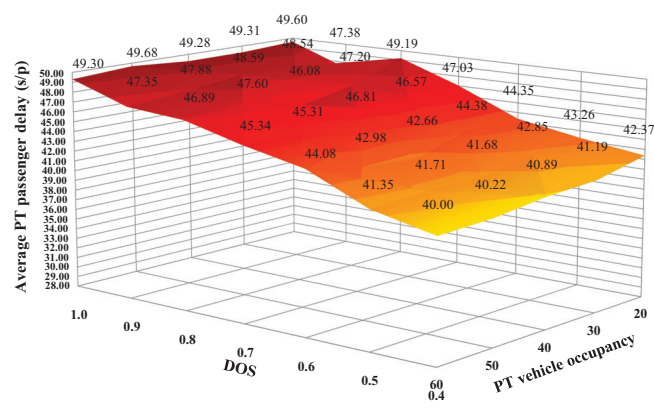
(a) Average passenger delay - SYNCHRO



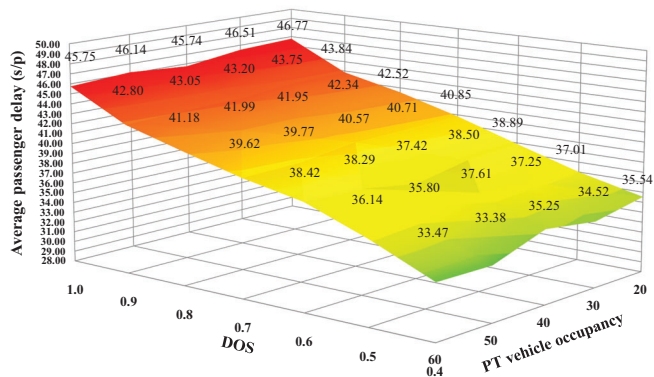
(b) Average PT passenger delay - SYNCHRO



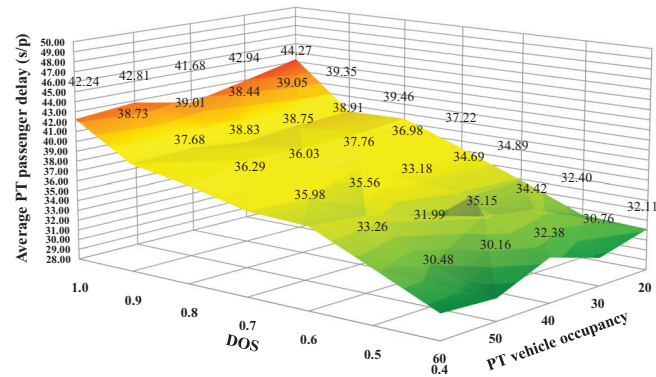
(c) Average passenger delay - KATC-1



(d) Average PT passenger delay - KATC-1



(e) Average passenger delay - KATC-2



(f) Average PT passenger delay - KATC-2

FIGURE 8 Results of the simulation experiments

The optimal solutions of the SYNCHRO model, an extensively used vehicle-based optimization method, are calculated for each of the DOSs and therefore their performance can be effectively evaluated against the passenger-based structure of the KATC models. The KATC-1 model with the capability of merely optimizing green time is considered as an alternative passenger-based method to provide a basis for investigating the effect of phase sequence optimization in the main model (KATC-2) on performance improvements. The cycle length for all methods is set to 120 s. Seven DOS levels and five PT vehicle

occupancy values are combined to develop 35 scenarios for the simulation experiments. Each scenario is run ten times in the VISSIM environment with an analysis horizon of 1 h and considering a 2-cycle warm-up interval.

5.4 | Results and discussion

The results of the average passenger delay and average PT passenger delay for the simulation experiments with optimal

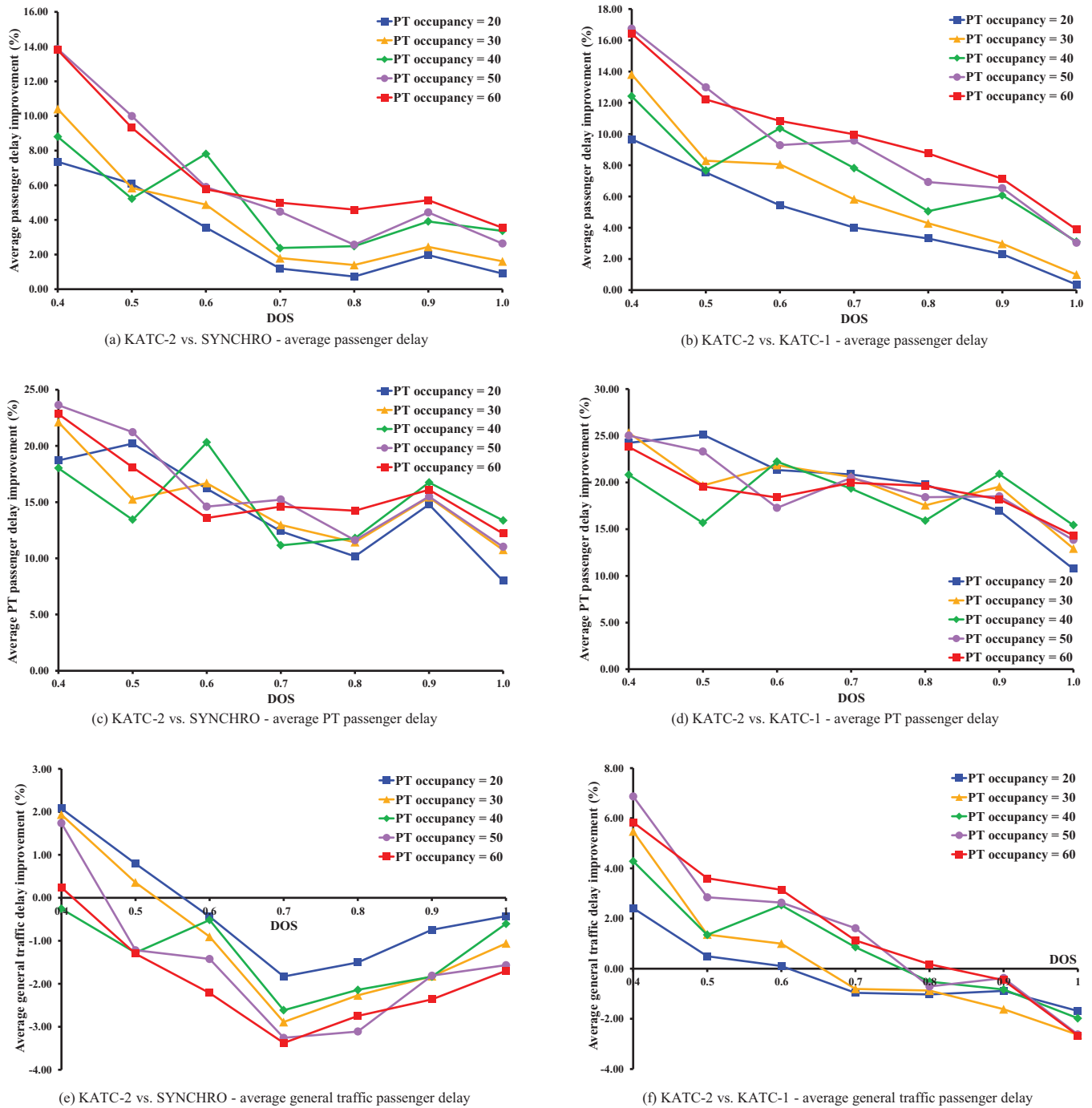


FIGURE 9 Delay improvements in KATC-2 scenarios compared to SYNCHRO and KATC-1 scenarios

solutions of SYNCHRO and KATC models are illustrated in Figure 8. The network results are averaged over ten simulation runs and the average values are illustrated as 3D surfaces. The results of SYNCHRO optimal solutions in Figure 8a,b indicate that average passenger delay and average PT passenger delay steadily increase with DOS. PT vehicle occupancy shows no significant effect on the respective results reflecting the vehicle-based structure of SYNCHRO. According to Figure 8c, the optimization of green time in the KATC-1 model reduces aver-

age passenger delay compared to the SYNCHRO model in 3 out of 35 scenarios, all occurring at DOS of 1.0. The results range from -0.6% to 5.9% change in average passenger delay relative to the SYNCHRO optimal solution. Regarding Figure 8d, SYNCHRO also demonstrates better performance in all scenarios compared to the KATC-1 model in terms of average PT passenger delay. However, the increase in average PT passenger delay compared to SYNCHRO is restricted to 12.0% in the extreme case. It can be inferred that under low congestion

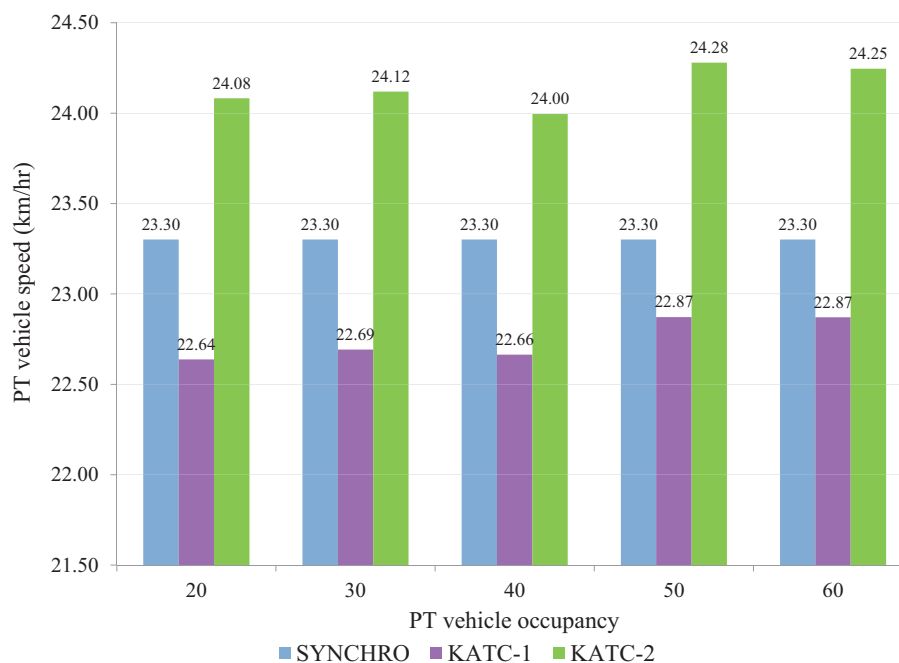


FIGURE 10 Average speed of PT vehicles for various vehicle occupancies (DOS = 0.7)

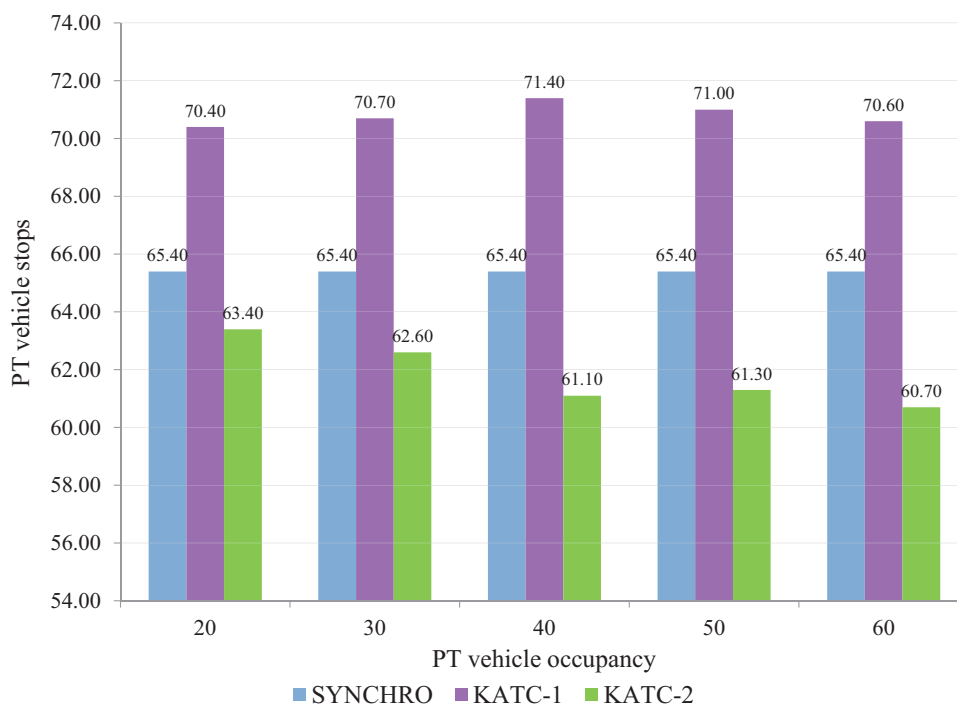


FIGURE 11 Number of stops of PT vehicles for various vehicle occupancies (DOS = 0.7)

levels and stable traffic demand conditions, the fixed signal timings delivered by the deterministic optimization in the SYNCHRO model seem to offer a more reliable performance than the solutions of GA-based stochastic optimization in KATC-1.

The reported results in Figure 8e explicitly indicate a significant improvement in delay experienced by passengers due to introducing the feature of phase sequence optimization in the KATC-2 model. As for the data presented in Figure 8f, higher-order benefits are achieved in the values of average PT